Sunny Zhang

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Education

University of Toronto, St. George 2025 – 2029
Honours BSc, Double Major in Computer Science and Mathematics *GPA: 3.7/4.0*

Publications

1. **Sunny Zhang**, Francesco Febbo, Harshit Saini. *Language–Action Decoupling: Speech as a Weak Proxy for Action in LLM Hierarchies*. sunnyzhang.dev. **Accepted to four NeurIPS 2026 workshops**: IAB, AIWILD, VerifyAgents, SocialAgent.

Experience

Founder & Lead Quantitative Researcher Jul 2026 – Present
UofT Decision Systems (UTDSYS) *Toronto, ON*

- Designed and pre-registered a study of robust mean-variance optimization that sizes the uncertainty set with a bootstrap frontier, tested against equal weighting and minimum variance over 53 years and 3 markets.
- Cut volatility 10–13% and drawdowns 6–7 pt vs. equal weighting in all markets ($p < 0.001$) at similar Sharpe.
- Deployed it to Alpaca paper trading with a scheduled monthly rebalancer; live since Nov 2026.

AI Research Intern Jun 2026 – Sep 2026
Algoverse AI Research Program *Remote*

- Led a 3-person team, came up with the architecture and implemented a 4-agent Llama-3.3-70B trading simulator in Python; ran 120 episodes that produced 2,400 labeled decisions.
- Built a seeded experiment harness for 5 pressure levels and 2 authority conditions; logged the messages and capital decisions separately to compare LLM language with its actions.
- Found managers relayed capital threats in 0.3% of messages sent down against 32.0% received as the trader capital gap grew from 0.55 to 0.85: speech carried the sign of an action, not its size.

Projects

Tiny Society | *Python, LLM Agents, FastAPI, Next.js*

- Built a full-stack simulator where 7 LLM agents with ranked memory, plans, and reflection interact.
- Developed a "World Lens" that holds a world's premise and reuses it in every agent prompt.

Feedback or Annealing? | *Python, NumPy, Numba, Matplotlib*

- Built a reproducible study of adaptive state aggregation for dynamic programming on MDPs: 3 width rules benchmarked on a 9,261-state inventory MDP across 20 shared seeds and 5 compute budgets.
- Showed plain value iteration beat every aggregation method: it converged in 4.2M backups while each aggregation run spent 9.2M+, and the residual feedback signal improved error by at most 0.2%.

QuantiFi 2026 Algorithmic Trading — 2nd in PnL of 30+ Teams | *Python, NumPy, Pandas*

- Led a 4-person team building a market-making engine for 26 products that trades against per-family fair values (jump regimes, option vol smile, cross-asset identity) and skews quotes against inventory.
- Guarded against overfitting with leave-one-day-out tuning, parameters chosen on wide plateaus rather than peaks, and only signals that held on all 3 days; 489k PnL, 478k under a stricter fill rule.

Technical Skills

Languages: Python, SQL, TypeScript, C++
Machine Learning: PyTorch, scikit-learn, Gymnasium, Hugging Face
Scientific Computing: NumPy, SciPy, Pandas, Numba, Matplotlib
Systems & Tools: FastAPI, Docker, Git, pytest, Supabase